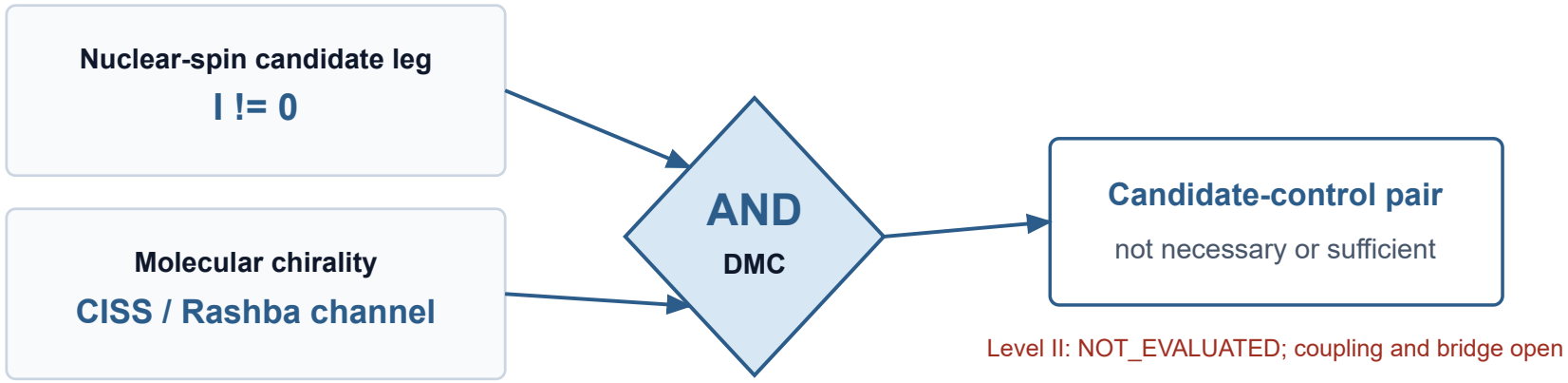
Figure V1.16.2

Dual Material Constraint candidate controls

Candidate controls: nuclear spin and molecular chirality; no Level-II verdict.



Substrate	I != 0 leg	chirality leg	DMC control pair	status / caveat
28Si digital control	☒ NO I = 0	☒ NO achiral lattice	**MISSING** legacy zero-rate scenario only	**NOT_EVALUATED** missing-leg control only
natural silicon	M MIXED 29Si low abundance	☒ NO achiral lattice	**MISSING** missing chiral spin architecture	**NOT_EVALUATED** no organized CISS channel
beta-tubulin Trp branch	Y YES I != 0 nuclei	Y YES L-chiral protein	**PRESENT** candidate controls only	**NOT_EVALUATED** legacy zero scenario; coupling not derived
NV + chiral h-BN cap	Y YES N spin register	Y YES chiral cap	**PRESENT** engineered surrogate	**NOT_EVALUATED** present; O.21 closure pending conditional eta_Zeno=26.8; Level II: NOT_EVALUATED

Control discipline: missing legs do not imply zero coupling or a Level-II verdict.